Web Technologies

.....AND SECURITY

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Chapter 6

JavaScript: Client-Side Scripting

Objectives

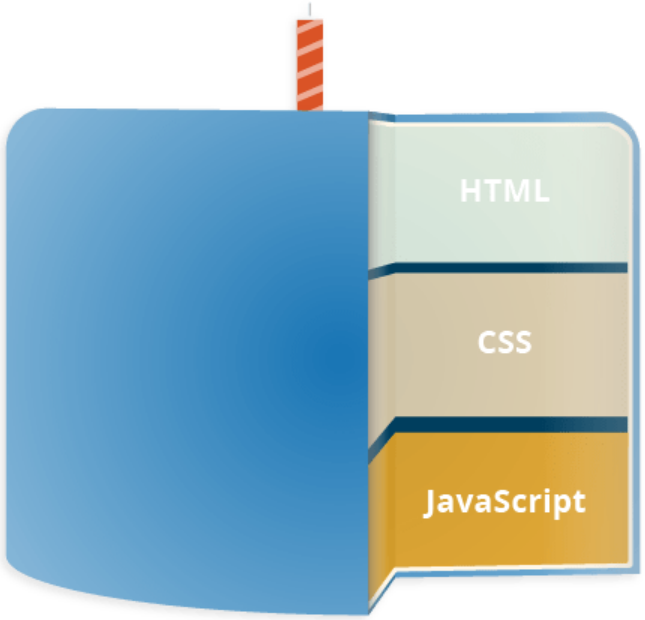
1. What is JavaScript
2. JavaScript Design
3. Using JavaScript
4. JavaScript Syntax
5. JavaScript Objects
6. The DOM
7. JavaScript Events
8. Forms

What Is JavaScript

- **JavaScript** is a scripting language that enables you to create dynamically updating content, control multimedia, animate images, and allows you to implement complex features on web pages
  - Every time a web page does more than just sit there and display static information for you to look at you can bet that JavaScript is probably involved.
    - e.g. displaying timely content updates, interactive maps, animated 2D/3D graphics, scrolling video jukeboxes, etc.
- With JavaScript we can animate, move, transition, hide, and show parts of the page rather than refresh an entire page from the server.
- **JavaScript** runs right inside the browser
- **JS** is object oriented in that almost everything in the language is an object

What Is JavaScript

- It is the third layer of the layer cake of standard web technologies,
  - The three layers build on top of one another nicely.
- **Prerequisites:** Basic computer literacy, a basic understanding of **HTML** and **CSS**.
- **Objective:** To gain familiarity with what JavaScript is, what it can do, and how it fits into a web site.



Why Study JavaScript?

- JavaScript is one of the 3 languages all web developers must learn:
  1. **HTML** to define the content of web pages
  2. **CSS** to specify the layout of web pages
  3. **JavaScript** to program the behaviour of web pages

Advantages Of Client-side Scripting

- Processing can be offloaded from the server to client machines, thereby reducing the load on the server.
- The browser can respond more rapidly to user events than a request to a remote server ever could, which improves the user experience.
- JavaScript can interact with the downloaded HTML in a way that the server cannot, creating a user experience more like desktop software than simple HTML ever could.

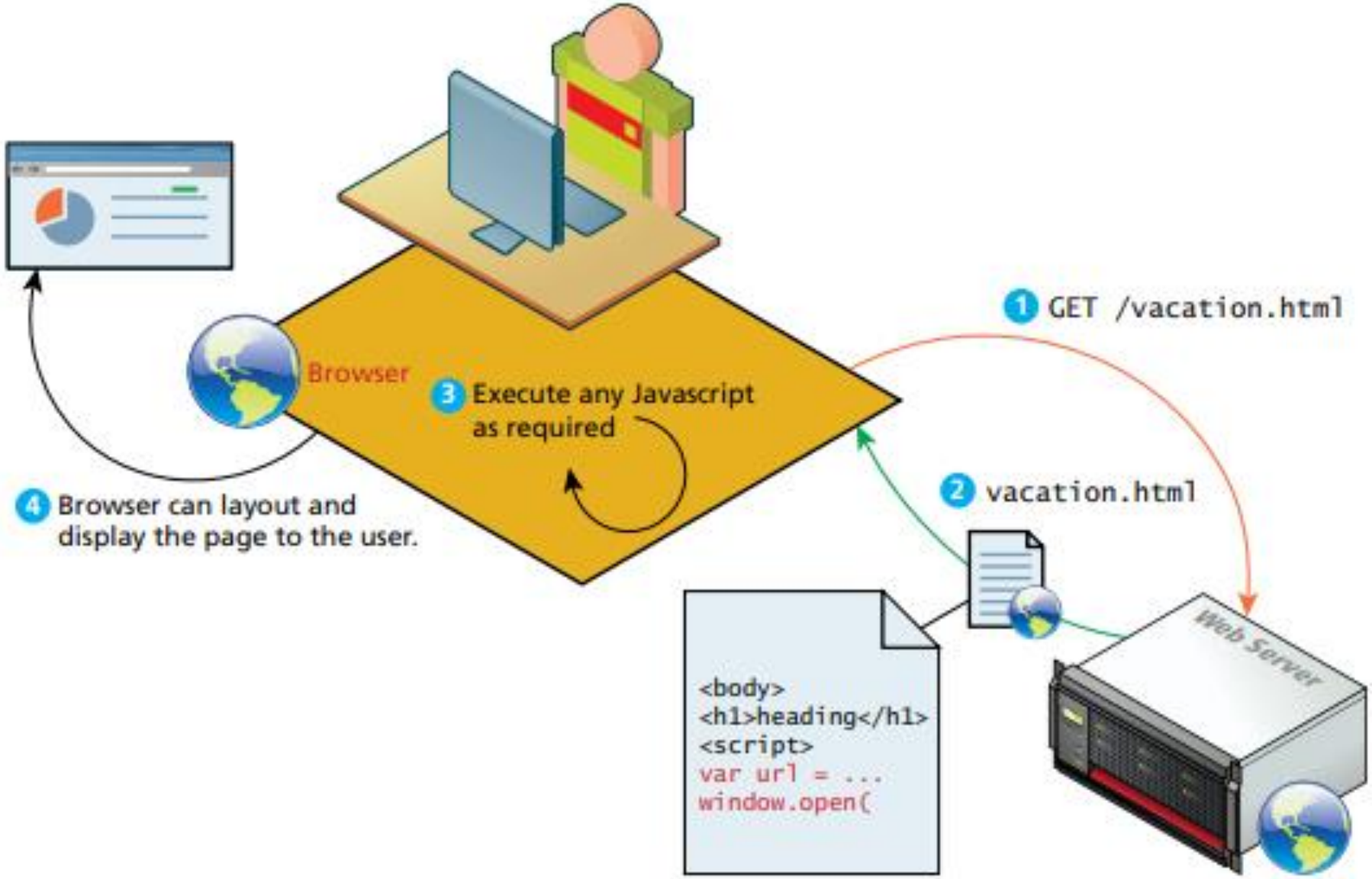


FIGURE 6.1 Downloading and executing a client-side JavaScript script

Disadvantages Of Client-side Scripting

- The disadvantages of client-side scripting are mostly related to how programmers use JavaScript in their applications.
  - There is no guarantee that the client has JavaScript enabled
    - Any required functionality must be housed on the server, despite the possibility that it could be offloaded.
  - The mode of behaviour between various browsers and operating systems make it difficult to test for all potential client configurations.
    - What works in one browser, may generate an error in another.
  - JavaScript-heavy web applications can be complicated to debug and maintain.
    - blending HTML and JavaScript together decreases code readability, and increases the difficulty of web development

Reading Task (Pages 280 – 285)

- JavaScript's History and Uses
- Synchronous and asynchronous data requests

JavaScript Design Principles

- **Layers:** In object-oriented programming, a software layer is a way of conceptually grouping programming classes that have similar functionality and dependencies.
- Common software design layer names include:
  - **Presentation layer:** Classes focused on the user interface.
  - **Business layer:** Classes that model real-world entities, such as customers, products, and sales.
  - **Data layer:** Classes that handle the interaction with the data sources.

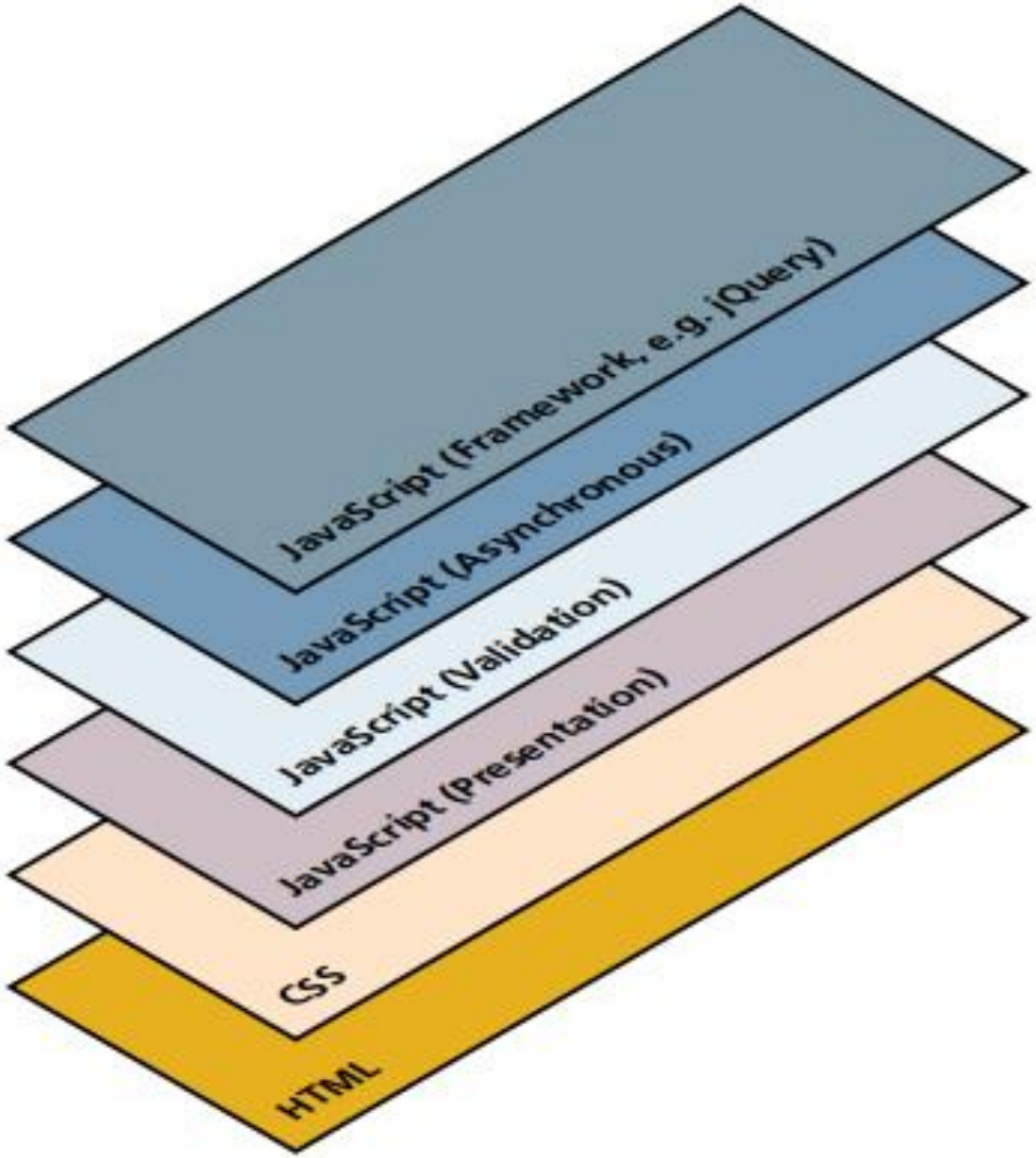


FIGURE 6.8 JavaScript layers

Presentation Layer

- This type of programming focuses on the display of information.
- JavaScript can alter the HTML of a page, which results in a change, visible to the user.
- These presentation layer applications include common things like creating, hiding, using tabs to show multiple views, or having arrows to page through result sets.
- This layer is most closely related to the user experience and the most visible to the end user.

Validation Layer

- JavaScript can be also used to validate logical aspects of the user's experience.
  - E.g. validating a form to make sure the email entered is valid before sending it along.
- It is often used in conjunction with the presentation layer to create a coherent user experience, where a message to the presentation layer highlights bad fields.
- Both layers exist on the client machine, although the intention is to pre-validate forms before making transmissions back to the server

Asynchronous Layers

- JavaScript normally operates in a **synchronous manner**
  - A request sent to the server requires a response before the next lines of code can be executed.
  - During the wait between request and response the browser sits in a loading state and only updates upon receiving the response.
- In contrast, an **asynchronous layer** can route requests to the server in the background.
  - As certain events are triggered, the **JS** sends the HTTP requests to the server, but while waiting for the response, the rest of the application functions normally
  - the browser isn't in a loading state.
  - When the response arrives JavaScript will (perhaps) update a portion of the page.
- Asynchronous layers are considered advanced versions of the presentation and validation layers above.

Users Without JavaScript

- A client may not have JavaScript because they are a web crawler, have a browser plug-in, are using a text browser, or are visually impaired.
  - **Web crawler:** A web crawler is a client running on behalf of a search engine to download your site, so that it can eventually be featured in their search results.
  - **Browser plug-in:** A browser plug-in is a piece of software that works within the browser, that might interfere with JavaScript.
  - **Text-based client**. Some clients are using a text- based browser.
    - Text-based browsers are widely deployed on web servers, which are often accessed using a command-line interface.
  - **Visually disabled client.** A visually disabled client will use special web browsing software to read the contents of a web page out loud to them.
    - These specialized browsers do not interpret JavaScript, and some JavaScript on sites is not accessible to these users.

```
Quick Connect   Profiles

#                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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      + Chapter 2
      + Chapter 3
      + Chapter 4
      + Chapter 5
      + Chapter 6
      + Chapter 7
      + Chapter 8
      + Chapter 9
      + Chapter 11
  * Lab Manuals
  * Presentations
      + Presentation 2
      + Presentation 3
  * Case Studies



Testimonials



Blog



Links
(NORMAL LINK)  Use right-arrow or <return> to activate.
 Arrow keys: Up and Down to move.  Right to follow a link; Left to go back.
 H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list

Connected to vulpix
```

FIGURE 6.9 Surfing the web with Lynx

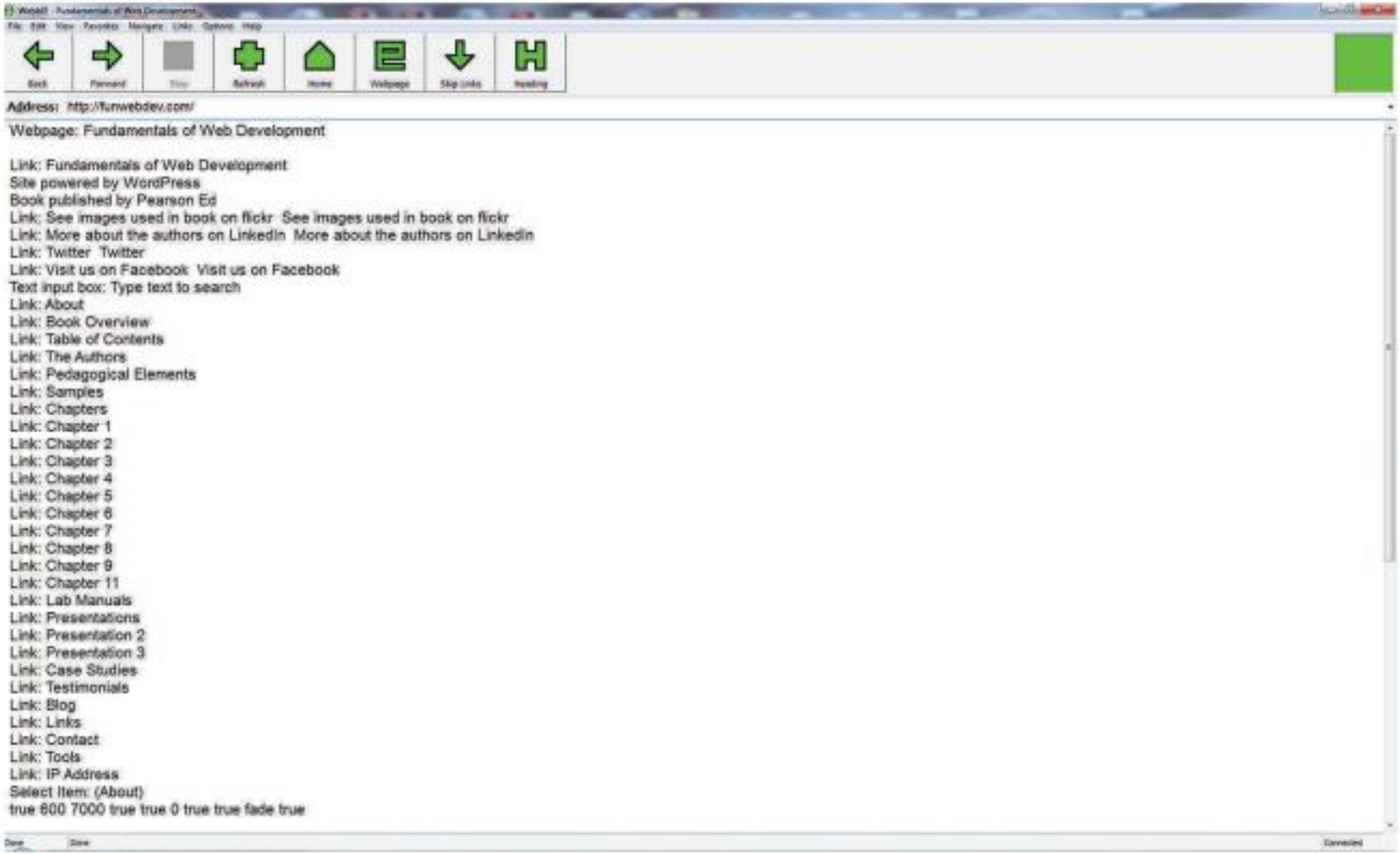


FIGURE 6.10 Screenshot of WebIE, browser for the visually impaired

Adding JavaScript into an HTML Document

- You can add JavaScript code in an HTML document by employing the dedicated HTML tag **<script>** that wraps around JavaScript code.
- The **<script>** tag can be placed in the **<head>** section of your HTML or in the **<body>** section, depending on when you want the JavaScript to load.
- JavaScript code can go inside of the document **<head>** section in order to keep them contained and out of the main content of your HTML document.
- Just as CSS, JavaScript can be linked to an HTML page in a number of ways/styles:
  - **Inline, Embedded, External**
- Just as with CSS these can be combined, but external is the preferred method for cleanliness and ease of maintenance

Inline JavaScript

- **Inline JavaScript** refers to the practice of including JavaScript code directly within certain HTML attributes
- This method is used when we have to call a function in the HTML event attributes.
  - E.g. adding JavaScript code directly eg., OnMover event, OnClick, etc.
  - Can add JavaScript directly in the html without using the **<script>.... </script>** tag.
- Inline JavaScript is a real maintenance nightmare, requiring maintainers to scan through almost every line of HTML looking for your inline JavaScript.
  - inline JavaScript is much worse than inline CSS

```
<a href="JavaScript:OpenWindow();"more info</a>
<input type="button" onclick="alert('Are you sure?');" />
```

LISTING 6.1 Inline JavaScript example

```html
<!DOCTYPE html >
<html>
<head>
<title> page title</title>
</head>
<body>
<p>
<a href="#" onClick="alert('Welcome !');">Click Me</a>
</p>
<p> In this example we saw how to use inline JavaScript or directly in an HTML
tag. </p>
</body>
</html>
```

Demo

Embedded JavaScript

- Embedded JavaScript refers to the practice of placing JavaScript code within a **<script>** element
- To add the JavaScript code into the HTML pages, web developer can use the **<script>.....</script>** tag of the HTML that wrap around JavaScript code inside the HTML program/page.
- Users can also define JavaScript code in the **<body>** tag (or we can say body section) or **<head>** tag because it completely depends on the structure of the web page that the users use.

```
<script type="text/javascript">
/* A JavaScript Comment */
alert ("Hello World!");
</script>
```

LISTING 6.2 Embedded JavaScript example

```html
<!DOCTYPE html >
<html>
<head>
<title> page title</title>

<script>
document.addEventListener('DOMContentLoaded', () => {
  function createParagraph() {
    const para = document.createElement('p');
    para.textContent = 'You clicked the button!';
    document.body.appendChild(para);
  }

  const buttons = document.querySelectorAll('button');

  for (const button of buttons) {
    button.addEventListener('click', createParagraph);
  }
});
</script>

</head>

<body>

  <button onclick="createParagraph()">Click me!</button>

<p> In this example we saw how to use embedded directly in an HTML document.</p>
</body>
</html>
```

Demo?

External JavaScript

- We can also create a separate file to hold the code of JavaScript with the **(.js)** extension and later incorporate/include it into our HTML document using the **src** attribute of the **<script>** tag.
- It becomes very helpful if we want to use the same code in multiple HTML documents.
- It also saves the developers from the task of writing the same code over and over again and makes it easier to maintain web pages

```
<head>
   <script type="text/JavaScript" src="greeting.js">
   </script>
</head>
```

LISTING 6.3 External JavaScript example

External JavaScript …

- First, create a new file in the same directory as your sample HTML file.
- Call it script.js make sure it has that .js filename extension, as that's how it is recognized as JavaScript.
- Replace your current <script> element with the following:
  - <**script** **src**="**script.js**"></**script**>
- Inside **script.js**, add the following script:
- Save and refresh your browser

```
function createParagraph() {
  const para = document.createElement('p');
  para.textContent = 'You clicked the button!';
  document.body.appendChild(para);
}

const buttons = document.querySelectorAll('button');

for (const button of buttons) {
  button.addEventListener('click', createParagraph);
}
```

SageCRMDev7Examples.js Sage CRM Cloud SData 2 Example.js

```
//These are examples
//Example Screen onload function
crm.ready(function () {
    //function script goes here
    var contextInfo = crm.getArgs();

    //add rule when company is in edit mode
    if (contextInfo.Act == "201" && contextInfo.Mode == "3") {
        //Add the onchange rule
        $("#comp_name").change(function () {
            re = /[^A-Za-z?&0-9.;:,\-() ]/;
            r = this.value.match(re);
            if (r) {
                crm.infoMessage(crm.getTrans('Errors', 'IllegalChars'));
                crm(this.name).background("#fff000");
                crm.hideButton('Save.gif');
            }
            else {
                crm.infoMessage(false);
                crm(this.name).background(false);
                crm.showButton('Save.gif');
            }
        });
    }
```

Benefits Of Using A Separate JavaScript

- In order to accommodate larger scripts or scripts that will be used across several pages, JavaScript code generally lives in one or more js files that are referenced within HTML documents, similarly to how external assets like CSS are referenced.
- The benefits of using a separate JavaScript file include:
  - Separating the HTML mark-up and JavaScript code to make both more straightforward
  - Separate files makes maintenance easier
  - When JavaScript files are cached, pages load more quickly

```
project/
├── css/
│   └── style.css
├── js/
│   └── script.js
└── index.html
```

Sample Code of CSS & JS in HTML

```html
<!DOCTYPE html>
<html lang="en-US">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Today's Date</title>
    <link rel="stylesheet" href="css/style.css">
</head>

<body>


<script src="js/script.js"></script>

</body>

</html>
```

JavaScript Syntax

- We will briefly cover the fundamental syntax for the most common programming constructs including
  - **variables,**
  - **assignment,**
  - **conditionals,**
  - **loops, and**
  - **arrays**
- We will also look at some advanced topics such as **events** and **classes.**

JavaScript's Reputation

- JavaScript's reputation for being unconventional not only stems from its strange way of implementing object-oriented principles but also from some odd syntax:
  - Everything is type sensitive, including function, class, and variable names.
    - E.g. online, Online, OnLine, and ONLINE are four distinct variable names
  - The scope of variables in blocks is not supported. This means variables declared inside a loop may be accessible outside of the loop, counter to what one would expect.
  - There is a **===** operator, which tests not only for equality but type equivalence.
  - Null and undefined are two distinctly different states for a variable.
  - Semicolons are not required, but are permitted (and encouraged).
  - There is no integer type, only number, which means floating-point rounding errors are prevalent even with values intended to be integers.

Variables

- Variables are containers for storing data (storing data values).
- Variables in JavaScript are dynamically typed, meaning a variable can be an integer, and then later a string, then later an object, if so desired.
- Assignment can happen at declaration-time by appending the value to the declaration, or at run time with a simple right-to-left assignment
- In a programming language, variables are used to store data values.
- JavaScript uses the keywords **var, let** and **const** to declare variables.
  - 4 Ways to Declare a JavaScript Variable: Using var, Using let, Using const, Using nothing
- An equal sign is used to assign values to variables.

Key pointers on Variable Usage: var or Const?

- **When to Use JavaScript var?**
  - The var keyword is used in all JavaScript code from 1995 to 2015.
  - The **let** and **const** keywords were added to JavaScript in 2015.
  - If you want your code to run in older browser, you must use var.
- **When to Use JavaScript const?**
  - If you want a general rule: always declare variables with const.
  - If you think the value of the variable can change, use let.

```html
<!DOCTYPE html>
<html>
<body>

<h1>JavaScript Variables</h1>

<p>In this example, price, price2, and total are variables.</p>

<p id="demo"></p>

<script>
const price1 = 5;
const price2 = 6;
let total = price1 + price2;
document.getElementById("demo").innerHTML =
"The total is: " + total;
</script>

</body>
</html>
```

```javascript
// How to create variables:
var x;
let y;

// How to use variables:
x = 5;
y = 6;
let z = x + y;
```

JavaScript Identifiers

- All JavaScript **variables** must be identified with unique names.
- These unique names are called **identifiers**.
- Identifiers can be short names (like x and y) or more descriptive names (age, sum, totalVolume).
- **The general rules for constructing names for variables (unique identifiers) are:**
  - Names can contain letters, digits, underscores, and dollar signs.
  - Names must begin with a letter
  - Names can also begin with $ and _
  - Names are case sensitive (y and Y are different variables)
  - Reserved words (like JavaScript keywords) cannot be used as names
- **JavaScript identifiers are case-sensitive.**

JavaScript Data Types

- Data types basically specify what kind of data can be stored and manipulated within a program.
- There are six basic data types in JavaScript which can be divided into three main categories:
  - **Primitive (or primary):** String, Number, and Boolean
  - **Composite (or reference):** Object, Array, and Function (which are all types of objects)
  - **Special data types:** Undefined and Null.
- Strings are written inside double or single quotes & Numbers are written without quotes.
  - If you put a number in quotes, it will be treated as a text string.
  - When adding a number and a string, JavaScript will treat the number as a string.
- JavaScript evaluates expressions from left to right.
  - Different sequences can produce different results

https://www.tutorialrepublic.com/javascript-tutorial/javascript-data-types.php

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript</h2>

<p>When adding a number and a string, JavaScript will treat the number as a
string.</p>

<p id="demo"></p>

<script>
let x = 16 + "Volvo";
document.getElementById("demo").innerHTML = x;
</script>

</body>
</html>
```

JavaScript

When adding a number and a string, JavaScript will treat the number as a string.

16Volvo

Left to right evaluation expressed

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript</h2>

<p>When adding a string and a number, JavaScript will treat the number as a
string.</p>

<p id="demo"></p>

<script>
let x = "Volvo" + 16;
document.getElementById("demo").innerHTML = x;
</script>

</body>
</html>
```

JavaScript

When adding a string and a number, JavaScript will treat the number as a string.

Volvo16

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript</h2>

<p>JavaScript evaluates expressions from left to right. Different sequences can
produce different results:</p>

<p id="demo"></p>

<!--In this example, JavaScript treats 16 and 4 as numbers, until it reaches
"Volvo". -->

<script>
let x = 16 + 4 + "Volvo";
document.getElementById("demo").innerHTML = x;
</script>

</body>
</html>
```

JavaScript

JavaScript evaluates expressions from left to right. Different sequences can produce different results:

20Volvo

Left to right evaluation expressed again

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript</h2>

<p>JavaScript evaluates expressions from left to right. Different sequences can
produce different results:</p>

<p id="demo"></p>

<!-- In this example, since the first operand is a string, all operands are
treated as strings. -->

<script>
let x = "Volvo" + 16 + 4;
document.getElementById("demo").innerHTML = x;
</script>

</body>
</html>
```

JavaScript

JavaScript evaluates expressions from left to right. Different sequences can produce different results:

Volvo164

JavaScript Types are Dynamic

- JavaScript has dynamic types.
  - This means that the same variable can be used to hold different data types
- If you re-declare a JavaScript variable declared with **var**, it will not lose its value.
  - You cannot re-declare a variable declared with **let** or **const**.

```
let carName = "Volvo";
let carName;
```

This will not work

```
var carName = "Volvo";
var carName;
```

This will work

```
let x;          // Now x is undefined
x = 5;          // Now x is a Number
x = "John";     // Now x is a String
```

```html
<!DOCTYPE html>
<html>
<body>

<h1>JavaScript Variables</h1>

<!--If you re-declare a JavaScript variable, it will not lose its value. -->

<p id="demo"></p>

<script>
var carName = "Volvo";
var carName;
document.getElementById("demo").innerHTML = carName;
</script>

</body>
</html>
```

JavaScript Variables

Volvo

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Data Types</h2>

<!-- JavaScript has dynamic types. This means that the same variable can be used
to hold different data types -->

<p id="demo"></p>

<script>
let x;          // Now x is undefined
x = 5;          // Now x is a Number
x = "John";     // Now x is a String

document.getElementById("demo").innerHTML = x;
</script>

</body>
</html>
```

JavaScript Data Types

John

Operator	Description	Matches (x=9)
==	Equals	(x==9) is true (x=="9") is true
===	Exactly equals, including type	(x=="9") is false (x===9) is true
< , >	Less than, greater than	(x<5) is false
<= , >=	Less than or equal, greater than or equal	(x<=9) is true
!=	Not equal	(4!=x) is true
!==	Not equal in either value or type	(x!=="9") is true (x!==9) is false

TABLE 6.1 Comparison Operators

Logical Operators

- Comparison operators are useful, but without being able to combine several together, their usefulness would be severely limited.
- JavaScript includes Boolean operators, which allow us to build complicated expressions.
- The Boolean operators **and, or, and not** and their truth tables are listed in Table 6.2.
- Syntactically they are represented with **&& (and), | | (or), and ! (not).**

A	B	A && B
T	T	T
T	F	F
F	T	F
F	F	F

AND Truth Table

A	B	A || B
T	T	T
T	F	T
F	T	T
F	F	F

OR Truth Table

A	! A
T	F
F	T

NOT Truth Table

TABLE 6.2 AND, OR, and NOT Truth Tables

Conditionals - If, else if, …, else

- JavaScript's syntax is almost identical to that of PHP, Java, or C when it comes to conditional structures such as if and if else statements.
- In this syntax the condition to test is contained within ( ) brackets with the body contained in { } blocks.

```
var hourOfDay;   // var to hold hour of day, set it later...
var greeting;   // var to hold the greeting message.
if (hourOfDay > 4 && hourOfDay < 12){
   // if statement with condition
   greeting =  "Good Morning";
}
else if (hourOfDay >= 12 && hourOfDay < 20){
   // optional else if
   greeting =  "Good Afternoon";
}
else{ // optional else branch
   greeting = "Good Evening";
}
```

LISTING 6.4 Conditional statement setting a variable based on the hour of the day

Loops

- Like conditionals, loops use the **( )** and **{ }** blocks to define the condition and the body of the loop.
- **While Loops:** loops normally initialize a loop control variable before the loop, use it in the condition, and modify it within the loop.
  - Be sure that the variables that make up the condition are updated inside the loop (or elsewhere) to avoid an infinite loop!
- **For loop:** combines the common components of a loop: **initialization, condition**, and **post-loop operation** into one statement.
  - This statement begins with the for keyword and has the components placed between **( )** brackets, semicolon (;) separated

```
for (var i = 0; i < 10; i++){
  //do something with i
}
```

FIGURE 6.15 For loop

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript For Loop</h2>

<p id="demo"></p>

<script>
let text = "";

for (let i = 0; i < 5; i++) {
  text += "The number is " + i + "<br>";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

```
var i=0;
while(i < 10){
 //do something with i
 i++;
}
```

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript While Loop</h2>

<p id="demo"></p>

<script>
let text = "";
let i = 0;
while (i < 10) {
  text += "<br>The number is " + i;
  i++;
}
document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

JS Functions

- **JS function** is a block of code designed to perform a particular task.
- **JS function** is executed when "something" invokes it (calls it).
- **JS function** is defined with the **function** keyword, followed by a **name**, followed by parentheses **()**.
- Function names can contain letters, digits, underscores, and dollar signs (same rules as variables).
- The parentheses may include parameter names separated by commas:
  - (parameter1, parameter2, ...)
- The code to be executed, by the function, is placed inside curly brackets: **{}**

JS Function Syntax

```
function name(parameter1, parameter2, parameter3) {
  // code to be executed
}
```

```
function myFunction(p1, p2) {
  return p1 * p2;   // The function returns the product of p1 and p2
}
```

JS Function Syntax

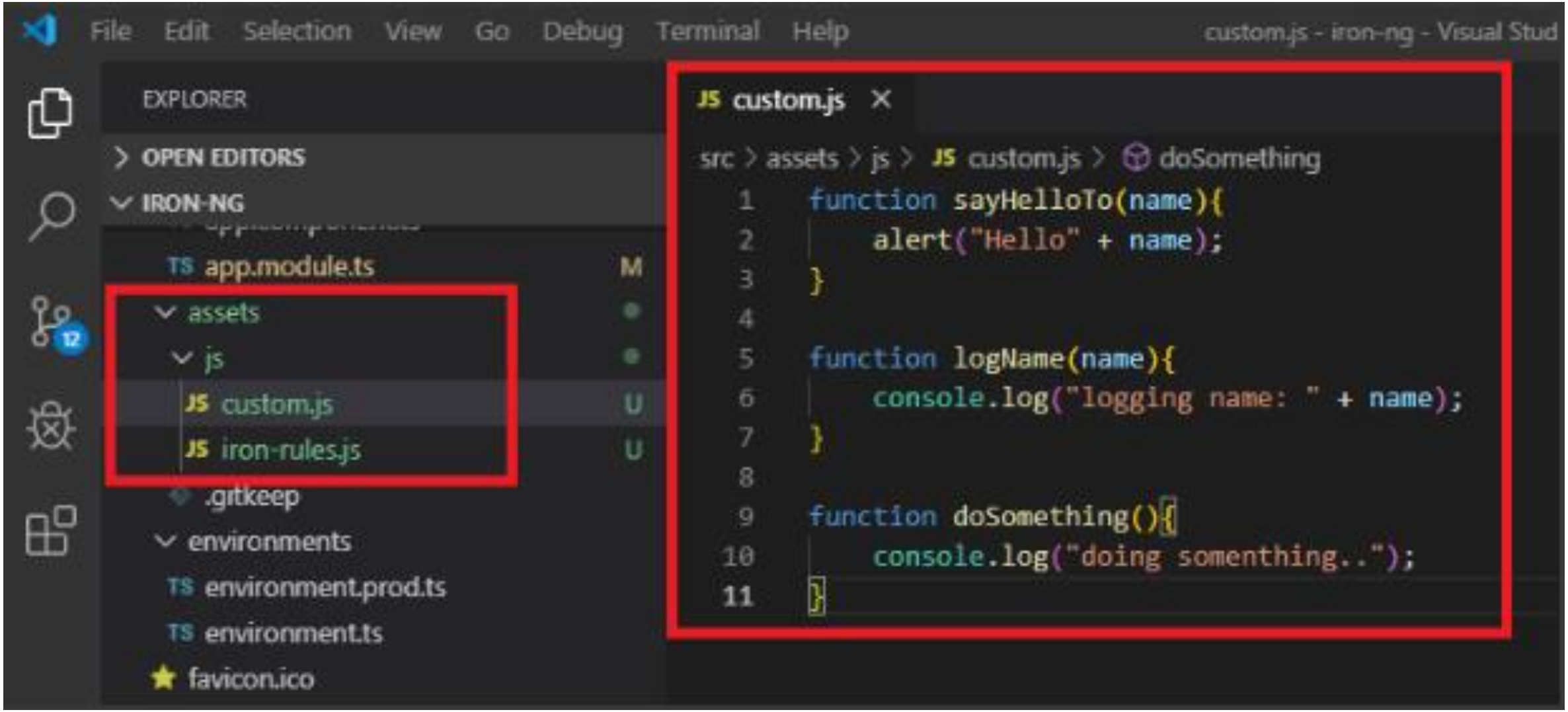
```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Functions</h2>

<!-- This example calls a function which performs a calculation, and returns the
result-->

<p id="demo"></p>

<script>
function myFunction(p1, p2) {
  return p1 * p2;
}
document.getElementById("demo").innerHTML = myFunction(4, 3);
</script>

</body>
</html>
```

```
function power(x,y){
   var pow=1;
   for (var i=0;i<y;i++){
       pow = pow*x;
   }
   return pow;
}
```

And called as

```
power(2,10);
```

Sample of JS file with functions



JS Functions

- Function parameters are listed inside the parentheses () in the function definition.
- Function arguments are the values received by the function when it is invoked.
- Inside the function, the arguments (the parameters) behave as local variables.
- **Since JavaScript is dynamically typed, functions do not require a return type, nor do the parameters require type.**
  - A function that lacks an explicit return statement will return **undefined**.
  - JS function does not need to return anything as far as your code goes. But as far as the JS engines are concerned: a function always returns something, be it explicitly via a return statement, or implicitly.
  - **If a function returns implicitly, its return value will always be undefined.**
- **Why Functions?**
  - You can reuse code: Define the code once, and use it many times.
  - You can use the same code many times with different arguments, to produce different results.

Alert

- JavaScript has three kind of popup boxes: **Alert box, Confirm box,** and **Prompt box**
- The **alert()** function makes the browser show a pop-up to the user, with whatever is passed being the message displayed.
- An alert box is often used if you want to make sure information comes through to the user.
- When an alert box pops up, the user will have to click "OK" to proceed.
  - The pop-up may appear different to each user depending on their browser configuration.
  - What is universal is that the pop-up obscures the underlying web page, and no actions can be done until the pop-up is dismissed.
- **Alerts are not used in production code, but are a useful tool for debugging and illustration purposes.**

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Alert</h2>

<button onclick="myFunction()">Try it</button>

<script>
function myFunction() {
  alert("I am an alert box!");
}
</script>

</body>
</html>
```

Errors Using try and catch

- When the browser's JavaScript engine encounters an error, it will throw an exception.
- These exceptions interrupt the regular, sequential execution of the program and can stop the JavaScript engine altogether.
  - You can optionally catch these errors preventing disruption of the program using the

    try – catch block

```
try {
  nonexistantfunction("hello");
}
catch(err) {
  alert("An exception was caught:" + err);
}
```

LISTING 6.5 Try-catch statement

Personalized Exceptions: throw

- Although **try-catch** can be used exclusively to catch built-in JavaScript errors, it can also be used by your programs, to throw your own messages.
- The **throw** keyword stops normal sequential execution, just like the built-in exceptions

```
try {
   var x = -1;
   if (x<0)
      throw "smallerthan0Error";
}
catch(err){
   alert (err + "was thrown");
}
```

LISTING 6.6 Throwing a user-defined exception

When to use Try-catch and throw statements

- Try-catch and throw statements should be used for **abnormal or exceptional** cases in your program.
- Throwing an exception disrupts the sequential execution of a program.
- When the exception is thrown all subsequent code is not executed until the catch statement is reached.
- This reinforces why try-catch is for exceptional cases.